CONCEPT NOTE

GridSense

A Self-Calibrating, Map-Aware Computer Vision System for Automated Traffic Violation Detection and Evidence Generation

Problem Statement 3: Automated Photo Identification and Classification for Traffic Violations Using Computer Vision

Submission type: Idea / Concept Note (no working model required, per Problem Statement 3 guidelines). External datasets referenced where applicable for the proposed evaluation strategy.

1. Problem Understanding

Traffic surveillance cameras across Indian cities generate an enormous volume of images every day, far beyond what manual review teams can realistically inspect. Manual inspection is slow, inconsistent across reviewers, and does not scale as camera networks expand. Beyond the volume problem, three structural issues limit today's automated systems:

- **Manual camera calibration —** every new camera or junction requires an engineer to physically define stop-lines, lane direction, and restricted zones in pixel coordinates. This does not scale to city-wide or state-wide deployment.
- **Rigid, hardcoded violation logic —** most systems use a separate hardcoded rule or model per violation type, making it expensive to add new violation categories or adapt to local traffic patterns.
- **Weak evidentiary value —** a single still photograph is often insufficient or contestable as legal evidence, particularly for judgment-based violations such as signal timing or stop-line crossing.

GridSense is designed specifically around solving these three structural gaps, in addition to the core computer vision tasks (detection, classification, OCR, analytics) listed in the problem statement.

2. Proposed Approach

GridSense is an AI-based traffic image analysis pipeline built on three core pillars, supported by a fourth evidentiary safeguard. Together, these allow the system to be deployed at any new junction with minimal manual configuration, to generalise to new violation types without retraining, and to produce evidence that holds up to scrutiny.

2.1 Pillar 1 — Map-Aware Auto-Calibration

Instead of manually marking stop-lines, lane directions, and no-parking zones on every camera feed, GridSense uses geo-referenced road network data (lane count, lane direction, junction geometry, and stop-line position) to automatically calibrate a new camera. Given the camera's location and orientation, a homography transformation maps real-world road geometry onto the camera's pixel space. A new camera can therefore become operational without any manual zone-drawing, which directly addresses the scalability requirement in the problem statement.

2.2 Pillar 2 — Scene-Graph Violation Reasoning

Rather than training a separate, hardcoded classifier for every violation type, GridSense represents each analysed frame (or short frame sequence) as a scene graph: vehicles, riders, pedestrians, the current signal state, and calibrated lane/zone regions are nodes, and their spatial and temporal relationships are edges. Violations are inferred by evaluating relationship patterns over this graph — for example, a rider node lacking an associated helmet-region node indicates a helmet violation, while a vehicle node whose motion vector opposes its lane-direction node indicates wrong-side driving. New violation types can be added by defining a new graph pattern, without retraining the underlying perception model.

2.3 Pillar 3 — Junction Memory (Adaptive Feedback Loop)

Every camera location has its own recurring sources of false positives — shadows, glare, partial occlusion from poles or trees, or unusual vehicle types. When a human reviewer rejects a flagged violation during the review stage, GridSense records this as site-specific feedback and locally adjusts the confidence threshold or masks the problematic region for that specific camera only. This allows a single global model to be deployed across many junctions while still adapting to local conditions, without needing to retrain or redeploy the entire model.

2.4 Evidentiary Safeguard — Multi-Frame Evidence Chain

Single still images are frequently contested in legal proceedings (for example, disputes over whether a signal was red or yellow, or whether a vehicle fully crossed a stop-line). Instead of saving one photograph, GridSense captures a short burst of frames bracketing the violation moment along with a logged signal-state timestamp, then cryptographically hashes the bundle to create a tamper-evident evidence packet. This produces a stronger evidentiary record than a single frame while remaining lightweight to store.

3. Methodology

The end-to-end pipeline maps directly onto the task list defined in the problem statement, as shown below.

Stage	What it does	Key techniques
1. Preprocessing	Normalises input images and corrects for low light, rain, shadows, and motion blur before detection.	CLAHE, low-light enhancement (Zero-DCE), classical de-rain/dehaze filters, deblurring
2. Detection & Tracking	Localises vehicles, riders, drivers, and pedestrians; classifies vehicle category; tracks objects across frame bursts.	Shared-backbone detector (YOLO/RT-DETR) with multiple task heads; ByteTrack for tracking
3. Map-Calibration	Aligns the camera's pixel space to real-world lane geometry, stop-lines, and restricted zones using geo-referenced map data.	Homography estimation from GPS/orientation + road network data
4. Violation Reasoning	Evaluates the scene graph against violation patterns to detect and classify each violation with a confidence score.	Scene-graph construction + rule/pattern evaluation over nodes and edges
5. Plate Recognition	Detects number plates and extracts registration text from the associated vehicle.	Small-object plate detector + OCR (e.g., PaddleOCR/CRNN) fine-tuned on Indian plate formats
6. Evidence Generation	Produces an annotated image/frame-burst, stores metadata and timestamps, and hashes the bundle for tamper-evidence.	Annotated overlays, structured metadata (JSON), hash-chained log
7. Review & Analytics	Routes low-confidence cases to human review, feeds rejections into Junction Memory, and generates violation statistics and trends.	Confidence-gated review queue, dashboards, heatmaps

A confidence-gated review step is included by design: violations above a high confidence threshold proceed to an auto-review queue, while borderline cases are routed for human verification. Fully automated issuance of fines without any human checkpoint is avoided, as this is both a fairness and legal-defensibility concern.

4. Assumptions

- Each camera has, or can be configured with, a known GPS location and approximate orientation, enabling the map-aware calibration step. Where this metadata is unavailable, orientation can be estimated from vanishing-point/road-edge cues in the image itself, with reduced calibration accuracy.
- A geo-referenced road network/lane-geometry data source (e.g., a mapping partner's data) is accessible for the camera's deployment region.
- Input images/frame bursts have a minimum resolution sufficient to resolve helmet, seatbelt, and license-plate regions (assumed ≥ 720p for near-field cameras).
- Signal-state information (red/yellow/green) is available either from the traffic signal controller or inferred visually from the signal head in-frame.
- Human reviewers remain part of the workflow for borderline-confidence cases; the system is positioned as a decision-support and evidence-generation tool rather than a fully autonomous enforcement system.
- Network/edge connectivity is sufficient to sync Junction Memory updates and evidence packets to a central store, with local buffering for intermittent connectivity.

5. Expected Evaluation Strategy

As permitted for Problem Statement 3, evaluation would draw on a combination of publicly available external datasets and synthetic/augmented data, since no proprietary dataset is mandated for this problem statement:

- Detection & vehicle classification — benchmarked against datasets such as the India Driving Dataset (IDD), which reflects unstructured Indian traffic conditions.
- Helmet / triple-riding / seatbelt classification — benchmarked against publicly available two-wheeler helmet-detection datasets, augmented with synthetic low-light/rain variants to test robustness.
- License plate OCR — benchmarked against publicly available Indian license-plate datasets, evaluated on full-plate and character-level accuracy.

Proposed metrics, mapped to each stage of the pipeline:

Component	Metric(s)	Target rationale
Vehicle/person detection	mAP@0.5, mAP@0.5:0.95	Standard detection benchmark, comparable across literature
Per-violation classification	Precision, Recall, F1-score, confusion matrix	Confusable classes (e.g., helmet vs. no-helmet under occlusion) need per-class visibility, not just overall accuracy

Map-calibration accuracy	Pixel-to-real-world alignment error (cm), vs. manual calibration baseline	Directly validates the core scalability claim
Junction Memory effectiveness	False-positive rate reduction over time, per camera	Validates that local adaptation reduces reviewer workload
License plate OCR	Character-level accuracy, full-plate match rate	Standard OCR evaluation practice
System efficiency	Frames/sec, end-to-end latency, cost per 1,000 images	Validates real-world scalability and deployment cost

Computational efficiency would additionally be assessed by benchmarking the shared-backbone detector on representative edge hardware (e.g., an NVIDIA Jetson-class device) against a cloud-GPU baseline, and by load-testing the inference API under simulated concurrent camera feeds to validate horizontal scalability.

6. Expected Impact

By removing manual per-camera calibration, GridSense is designed to scale from a single pilot junction to a city-wide or state-wide network with minimal incremental engineering effort. The scene-graph reasoning layer keeps the system extensible as new violation categories or local traffic regulations are introduced, and the multi-frame evidence chain strengthens the legal defensibility of automated enforcement — directly reducing manual reviewer workload while improving consistency, which is the central outcome sought by this problem statement.

7. Conclusion

GridSense reframes automated traffic violation detection as a calibration and reasoning problem, not just a detection problem. The combination of map-aware auto-calibration, scene-graph violation reasoning, and a site-level adaptive feedback loop addresses the scalability and maintainability gaps that limit today's deployed systems, while a multi-frame tamper-evident evidence chain addresses the evidentiary weaknesses of single-photo enforcement. The proposed approach is deliberately modular, allowing each component to be evaluated, piloted, and scaled independently.